

Woojin Bae

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RESEARCH INTERESTS

Design and Fabrication of Nanomaterials through Advanced Techniques

- **Nanomaterials:** Perovskite, Zeolite, Metal Oxide, Nanoparticle.
- **Advanced Techniques:** Transmission Electron Microscopy, Autonomous Laboratories.

EDUCATION

Seoul National University (SNU), Seoul, South Korea

Expected Aug. 2026

B.S. Chemical and Biological Engineering (CBE), *Cum Laude (expected)*

- GPA: 3.72/4.3 (overall), 3.67/4.0 (converted)
- Graduate Course: Topics in Inorganic Material and Semiconductor Processes (Fall 2025)

North Carolina State University (NCSU), Raleigh, NC

May 2025

Exchange student in Chemical and Biomolecular Engineering

- GPA: 4.0/4.0, AIChE at NC State

Daejeon Science High School for the Gifted, Daejeon, South Korea

Feb. 2020

High School for gifted students in science and mathematics

RESEARCH EXPERIENCE

Multi-dimensional Materials Chemistry Lab, SNU

Mar. 2024 – Present

Undergraduate Researcher - (PI: Prof. *Jungwon Park*)

Seoul, Korea

- Developed a DBSCAN-based protocol for quantitative analysis of SEI; SEI Structures Dependent Li-Morphology Characterized by cryo-TEM.
- Initiated and led a **collaborative project** across SNU and NCSU to optimize AI-driven synthesis of Mn-doped CsPb(Cl/Br)₃ QDs, through TEM analysis and DFT calculation.

Summer Undergraduate Initiative (SURI), ASU

May 2025 – Jul. 2025

Undergraduate Researcher - (PI: Prof. *Sandhya Susarla*)

Tempe, AZ

- Selected for the **SURI fellowship** at Arizona State University (\$5000 stipend).
- Synthesized perovskite nanocrystals with controlled morphologies to investigate phonon–polariton coupling with hexagonal boron nitride (hBN).
- Presented a poster at Summer Research Initiative Final Research Showcase.

Self-Driving Fluidic Lab, NCSU

Jan. 2025 – May 2025

Undergraduate Researcher - (PI: Prof. *Milad Abolhasani*)

Raleigh, NC

- Achieved Photoluminescence Quantum Yield (PLQY) improvement of Mn-doped CsPb(Cl/Br)₃ perovskite through genetic algorithm-based synthesis optimization in autonomous lab.
- Presented a poster at Spring Undergraduate Research & Creativity Symposium.

Center for Cell-Encapsulation Research, KAIST

Mar. 2018 – May 2019

Highschool R&E Researcher - (PI: Prof. *Insung Choi*)

Daejeon, Korea

- Created a porous titanium oxide shell to facilitate the in-vitro culturing of astrocytes.
- Confirmed the function of encapsulated astrocytes by analyzing the length of co-cultured neurons.

PUBLICATION / PRESENTATION

Jinge Xu.; **Bae, W.**; Milad Abolhasani. *, Autonomous Mn-doped Perovskite Nanocrystals synthesis by a Self-Driving lab. *Manuscript in preparation*

Bae, W.; Jung, S.; Ji, S.; Jeon, Y.; Park, J. *, SEI Structures Dependent Li-Morphology Characterized by Cryo-TEM. *The 2024 Annual Fall Conference of Korean Society of Microscopy*, Gyeongju, Korea, Nov 25-26, 2024.

Bae, W.; Lee, K.; Lee, C.; Kim, T.; Choi, I. *, Regulation of astrocyte growth using porous titanium oxide shells. *Korean Chemical Society Future Chemist Research Presentation*, Daegu, Korea, Oct 19, 2018.

Bae, W.; Lee, K.; Lee, C. *, Wettability Change of Self-Assembled Monolayers (SAMs) of Thiol on Gold. *The 51st Joint School Science Exhibition*, Hong Kong, Hong Kong, Aug 22-27, 2018.

HONORS & AWARDS

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|---|--------------------|
| Merit Award Korea Process Simulation Olympiad 2025, KICChE, Korea | Sep. 2025 |
| <ul style="list-style-type: none"> • Topic: Conducted feasibility study of Methanol synthesis process with Aveva process simulation (APS) (PI: Prof. <i>Jongmin Lee</i>) | |
| Research Grant Undergraduate Research Support Program, SNU College, SNU | May 2025 |
| <ul style="list-style-type: none"> • Selected for a \$2000 Research Grant for a collaborative project across SNU and NCSU. | |
| Excellence Research Award 2024 Student-Directed Education Program Symposium | Jan. 2025 |
| <ul style="list-style-type: none"> • Outstanding research on SEI structures dependent Li-morphology characterized by Cryo-TEM. | |
| Korea-U.S. STEM Exchange Scholarship Ministry of Trade, Industry and Energy, Korea | Nov. 2024 |
| <ul style="list-style-type: none"> • Awarded national \$9000 scholarship as a STEM-specialized exchange student. | |
| Excellence Award Undergraduate Research Program for Advanced Equipment, SNU | Jan. 2024 |
| <ul style="list-style-type: none"> • Investigated the relationship between microscopic and macroscopic morphology of hollow ZSM-5 under various synthesis conditions, utilizing TEM, SEM, and XRD. | |
| Team Section Winner We Change SNU Contest, SNU | Jul. 2022 |
| <ul style="list-style-type: none"> • A spatial-restricted messenger allowing messages to be accessed only at SNUfestival locations designated by the sender. | |
| The Education and Research Foundation Scholarship College of Engineering, SNU | Spring 2021 |
| 2020 Future Chemistry Talent Award KUCST, The Korean Chemical Society | Feb. 2020 |
| <ul style="list-style-type: none"> • Selected as one of the top 6 science high school students nationwide. | |
| POSTECH Presidential Award R&E Joint Symposium for Gifted High School Students | Jan. 2019 |
| <ul style="list-style-type: none"> • Awarded for outstanding poster presentation on astrocyte growth regulation using porous titanium oxide shells. | |

LEADERSHIP

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| Tokyo Forum Youth Session The University of Tokyo, Japan | Fall 2024 |
| <ul style="list-style-type: none"> • Representative of Korea, in a session on Aging Society and Low Birth Rate. • Participated in the wrap-up session hosted by the President of the University of Tokyo. | |
| HKUST Entrepreneurship Bootcamp Hong Kong University of Science and Technology | Jul. 2024 |
| <ul style="list-style-type: none"> • Led a global team in a mock-up competition and won third prize for developing a worldwide healthcare platform among international participants. | |
| Marketing manager SNUfestival, Student council organization, SNU | Jun. 2021 – May 2022 |
| <ul style="list-style-type: none"> • Promoted the festival through the online website and secured \$2200 sponsorship from a shared scooter company during the first in-person festival after the COVID-19 pandemic. | |
| Vice president CBE Interim Leadership, SNU | Dec. 2020 – Mar. 2021 |
| <ul style="list-style-type: none"> • Developed an online program for incoming freshman during the COVID-19 pandemic. | |

MEMBERSHIP & ACTIVITIES

Undergraduate Member | Korean-American Scientists and Engineers Association **Jan. 2025 – Present**

- A non-profit organization that fosters collaboration among Korean-American scientists and engineers. Delivered a research presentation at **Katalyst 2025**, a nationwide Korean-American undergraduate conference.

Young Engineers Honor Society | National Academy of Engineering of Korea **Nov. 2024 – Present**

- Korean engineering honor society under National Academy of Engineering of Korea.

SNU-UMN Culture Exchange Program | The University of Minnesota, Minneapolis, MN **Mar. 2024**

Winter Vacation Data Science Bootcamp: **Computing** | Graduate School of Data Science, SNU **Feb. 2024**

Republic of Korea Air Force | 11th Fighter Wing, Daegu, Korea **Aug. 2022 – May 2024**

- F-15K Slam Eagle Squadron. Selected as an **outstanding soldier**.

SNU in EU, Understanding European Union | Brussels School of Governance, Belgium **Jul. 2022**

- Study Abroad Program, delivered a presentation titled "The European External Action Service".

Kansas Academy of Mathematics and Science | Fort Hays State University, Hays, KS **Jul. 2018**

- Synthesized ZnSe quantum dots and applying them on Dye-sensitized solar cell. (PI: Prof. *Arvin J. Cruz*)

TEACHING & MENTORING EXPERIENCES

Korean Language Peer Mentor | Language Education Institute, SNU **Fall 2024**

Peer Tutor | Basic Computing: First Adventures in Computing, SNU **Fall 2024, Fall 2025**

Lecturer, TEM Seminar | Multi-dimensional Materials Chemistry Lab, SNU **May 2024**

Student Mentor, SNU Mentoring | Seoul National University Social Responsibility **Apr. 2023 – Present**

- Provided career guidance to the students from underserved communities, selected as an **outstanding mentor**.

TECHNICAL SKILLS & LANGUAGES

- **Instruments:** TEM, SEM, UV-Vis, OM, PL
- **Programming Languages:** Python, MATLAB, C
- **Software:** Gatan DigitalMicrograph, VASP, APS, Git, Adobe, Fusion 360
- **Languages:** Korean (Native), English (C1, TOEFL 103)